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Reference charts for weight gain and body mass index during pregnancy obtained from a healthy cohort

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Abstract

Objective. To create reference charts for weight gain and body mass index (BMI) in pregnancy derived from longitudinal data obtained in a representative sample of the Argentinean population.

Methods. A prospective cohort of 1439 healthy pregnant women was selected from antenatal clinics in seven different urban regions in Argentina. Serial anthropometric measurements were made at weeks 12, 16, 20, 24, 28, 32, 36 and in the last pre-natal control. Centile curves of body weight and BMI by gestational age were developed using the LMS method.

Results. Mean weight gain at 38 weeks of gestation was 11.9 ± 4.4 kg. There were no differences in total weight gain between women who enter pregnancy with low, normal or overweight; only those women with a pre-pregnancy BMI in the range of obesity showed a significantly lower weight gain (10.2 ± 4.8 kg). At 12 weeks of pregnancy, BMI values of the 10th, 50th and 90th centiles were 19.3, 22.8 and 29.0, and at 38 weeks these values were 23.3, 27.4 and 33.8, respectively.

Conclusion. This BMI for gestational age chart, based on women who delivered normal birth weight infants and processed with modern statistical methods, represents an improvement in pre-natal care monitoring.

Keywords: *Weight gain, pregnancy, body mass index, birth weight*

Introduction

Maternal weight gain monitoring is a key component of pre-natal care; either low or high weight gain during pregnancy can negatively affect maternal and fetal outcomes. The relationship between low gains and low birth weight, intrauterine growth retardation and later risk of several metabolic diseases has been well documented [1–4]. However, high weight gains have been associated with a greater risk of macrosomia, cesarean delivery and post-partum weight retention [5–7]. Although gestational weight gain charts have been used in clinical practice for many years, consensus about optimal weight gain during pregnancy is still controversial and many of the available instruments currently in use are questioned [8,9].

The American College of Obstetrics and Gynecology adapted the recommendations of the Institute of Medicine (IOM) and suggested a total weight gain between 12.7 and 18.2 kg for under-weight women, 11.4 and 15.9 kg for normal weight woman, 6.8 and 11.4 kg for overweight women and 6.8 kg for obese women [10,11]. Expressing weight gain in absolute values does not consider proportionality of recommended weight gain for short and tall women.

In Latin American countries the Rosso and Mardones chart, which enables the identification of mothers at risk of delivering either small or large for gestational age babies, has been widely adopted since 1987 [12,13]. Nevertheless, in the last decade this instrument has also been questioned because of its low predictive value of birth weight, and because it overestimates obesity [14,15].

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The aim of the present study was to create reference charts for weight gain and body mass index (BMI) in pregnancy derived from longitudinal data obtained in a Latin American population.

The World Health Organization recommends that a gestational weight gain chart should be based on longitudinal studies in a selected population with a low prevalence of maternal and fetal complications, with anthropometric measures before and during the course of pregnancy. Ideally, upper and lower limits around the average curve should be set, and BMI reference curve has the advantage that it takes into account the woman height during the whole course of pregnancy [16].

Population and methods

Population

A prospective cohort of 1439 pregnant women was selected from antenatal clinics in seven different urban regions in Argentina. Data were collected from May 2005 to December 2006.

Sample size was estimated taking into account a national prevalence of low birth weight of 7%, with a power of 0.80 and a significance level of 0.05, considering 25% of possible drop-outs.

The study protocol was conducted in agreement with local ethical standards according to Helsinki Declaration; all the subjects signed an informative consent form. The National Commission for Health Research approved and funded the project.

To be eligible for this study women had to be 19- to 49-years old, with singleton pregnancies at a gestational age less than 12 weeks at entry (or less than 16 weeks, if pre-pregnancy weight was accurately remembered), without concomitant pathologies, parity 0–5, non-smokers or smoking less than five cigarettes per day and no alcohol consumers or drinking less than 20 g ethanol per day.

Measures

At the first visit, a standard questionnaire on social characteristics and a medical history was completed. Gestational age was determined by dating the last menstrual period at the time of registration and corrected by first trimester ultrasonographic examinations if the difference exceeded 5 days. Pre-pregnancy weight was determined by maternal recall. Serial anthropometric measurements were made on calibrated scales at weeks 12, 16, 20, 24, 28, 32, 36 and in the last pre-natal control. BMI was calculated by dividing the maternal weight in kilograms by the square of the height in meters. Before data collection, a standardisation procedure was carried out to homogenise anthropometric results between investigators.

Infant birth weight, height and gestational age were recorded from the neonate clinical history.

Validation study

An independent sample of 80 clinical histories in each of the seven study areas, thus totalising 560 pregnancies, was collected for this purpose. Sample was intentionally biased to collect a higher than average number of low birth weight infants. Age, pre-pregnancy weight, height and at least three body weight determinations with accurate gestational age along pregnancy were recorded, as well as the newborn weight.

Data analysis

Centile curves of body weight and BMI by gestational age were developed for those pregnant women who delivered neonates with birth weights between 2500 and 4000 g. The LMS method that summarises the distribution by three curves representing the median (*M*), coefficient of variation (*S*) and skewness expressed as a Box-Cox power transform (*L*) was used to fit the smoothed curves (LMS ChartMaker, ©1997-2005 Medical Research Council, UK) [17].

Weight velocity during pregnancy was analysed using the individual delta-weights every 4-weeks intervals, standardising the mid-point of the interval by interpolation. The means and 95% confidence intervals of weight increments were then calculated for each 4-weeks interval.

Summary statistics were calculated using Epi-Info software, version 3.2. Quantitative data were expressed as means $\pm$ standard deviation (SDs) or as medians and 25th–75th centiles, according to the variable distribution. Quantitative data were analysed by contrasting means and qualitative data were analysed by using the chi-square test. A significance level of $p < 0.05$ was used in all tests.

In the validation study, receiver operating characteristic curves were used to define the limits of BMI associated with the likelihood ratios that best predict low or high birth weights.

Results

A cohort of 1439 pregnant women was initially recruited in the seven areas; 349 of them (24.2%) were lost to follow-up for different reasons (mainly failure to attend programmed visits or moving before delivery). Results are based on the 1090 pregnancies followed until delivery. There were no significant differences between those who completed the follow-up and those who not in age, income, percentage with a formal job, height or initial BMI. The number of previous pregnancies was different, with a higher frequency of first pregnancies in those who completed the follow-up (31.3 vs. 16.1%, chi-square test 28.39, $p = 0.00001$). Most pregnant women included in the charts (92.7%) attained five or more visits, and 68.9% completed the projected visit schedule of seven or more visits.

Characteristics of the subjects are presented in Table I.

In previous pregnancies, 4.1% of women had antecedents of diabetes, 2.6% reported pre-eclampsia and 1.4% eclampsia; and 7.3% had antecedents of hypertension.

During the follow-up, 1.93% developed diabetes, and 2.57% pre-eclampsia; there were no differences in maternal weight gain with those mothers who had no pathology during pregnancy (t-test 1.89, $p = 0.06$, and t-test 1.33, $p = 0.18$, respectively).

Anthropometric characteristics

The distribution of pre-pregnancy BMI, according to the categories defined by the IOM, is shown in Table II.

The median of weight gain at 38 weeks of gestation was 11.4 kg (inter-quartile interval 9–14.5 kg). Mean weight gain by pre-pregnancy BMI category is also shown in Table II. There were no differences in weight gain between women who enter pregnancy with low, normal or overweight; only those women with a pre-pregnancy BMI in the range of obesity showed a significantly lower weight gain.

Newborn characteristics

Birth outcomes refer to 1090 newborns, 51.6% were females, 5.7% were pre-term and 1.1% post-term, 96.2% presented adequate weight for gestational age and 0.9% presented minor birth defects. Other newborn characteristics are shown in Table III.

Birth weight was positively associated with pre-pregnancy BMI and with total maternal weight gain (adjusted R-squared: 0.133, $p = 0.0001$).

Weight for gestational age

LMS curves of body weight for gestational age were calculated for only those women who delivered newborns with a birth weight between 2500 and 4000 g, using 7095 determinations along pregnancy (Figure 1). Two outlier points (± 3 SD) were eliminated from the calculation. Specifications of the model that provided the best fit to generate the curves were:

No age power transformation; df $(M) = 7$; df $(S) = 4$ and df $(L) = 3$.

BMI for gestational age

LMS curves of BMI for gestational age were calculated using 7092 BMI determinations along pregnancy of only those women who delivered newborns with a birth weight between 2500 and 4000 g (Figure 2 and Table IV). Five outlier points

Table I. Characteristics of the pregnant women.

Age [years, mean (SD)]	27.0 (5.8)
Pre-pregnancy weight [kg, mean (SD)]	60.1 (12.5)
Height [cm, mean (SD)]	159 (7)
Pre-pregnancy BMI [kg/m^2, mean (SD)]	23.4 (4.3)
Parity [mean (SD)]	1.34 (1.2)
Nulliparae [N (%)]	341 (31.3)
Gestational age at entry [weeks, mean (SD)]	10.2 (2.3)
Household income below the second decile [N (%)]	119 (11)
Housewives [N (%)]	627 (58)
Pregnant women with health insurance [N (%)]	479 (44)
Education	
Incomplete primary school [N (%)]	28 (2.6)
Complete primary school [N (%)]	131 (12.1)
Incomplete secondary school [N (%)]	248 (22.9)
Completed secondary school [N (%)]	406 (37.4)
University [N (%)]	271 (25.0)
Smokers (less than 5 cigarettes/day) [N (%)]	101 (10.8)
Alcohol consumers (less than 20 g ethanol/day) [N (%)]	69 (7.3)

Table II. Pre-pregnancy BMI and weight gain at 38 weeks of gestation.

BMI category (IOM) (kg/m^2)	Frequency	%	Weight gain (kg) at 38 weeks of gestation (mean $\pm$ SD)
<19.8	161	14.8	12.2 ± 3.9
19.8–26	693	63.6	12.1 ± 4.3
>26–29	124	11.4	12.1 ± 5.0
>29	112	10.3	10.2 ± 4.8*
Total	1090		11.9 ± 4.4

*Significantly different from all others.

Table III. Newborn anthropometric characteristics.

Condition	
Mean birth weight (g)	3237 ± 490
Mean birth length (cm)	47.5 ± 3.8
Low birth weight (<2500 g)	5.05%
Insufficient birth weight (2500–2999 g)	21.28%
Normal birth weight (3000–4000 g)	68.99%
High birth weight (>4000 g)	4.68%

(± 3 SD) were eliminated from the calculation. Specifications of the model that provided the best fit to generate the curves were:

No age power transformation; df $(M) = 7$; df $(S) = 3$ and df $(L) = 3$.

BMI data are presented in z-score units in Figure 2 because the cut-off points that best predict birth weight were ± 1 SD in the validation sample.

Weight velocity

Weight increments were calculated for those women who gave birth to infants of normal birth weight (Table V), and compared with the weight increments of women who delivered infants below 3000 g. There were significant differences between these groups in the 20–24 weeks interval and in the 32–36 weeks interval.

Validation

BMI data from the auxiliary sample were converted to z-scores using the LMS values generated in the study. Low birth weight was best predicted by a BMI less than -1 SD (sensitivity: 37.43% and specificity: 84.17%). Prediction of high birth weight by a BMI

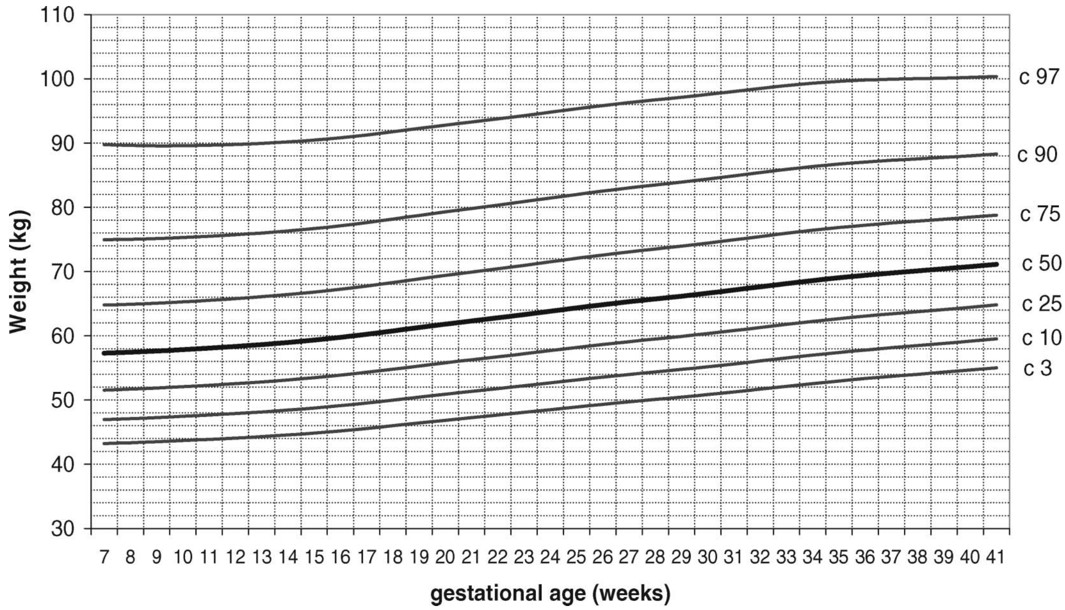

Figure 1. Body weight centile curves by gestational age.

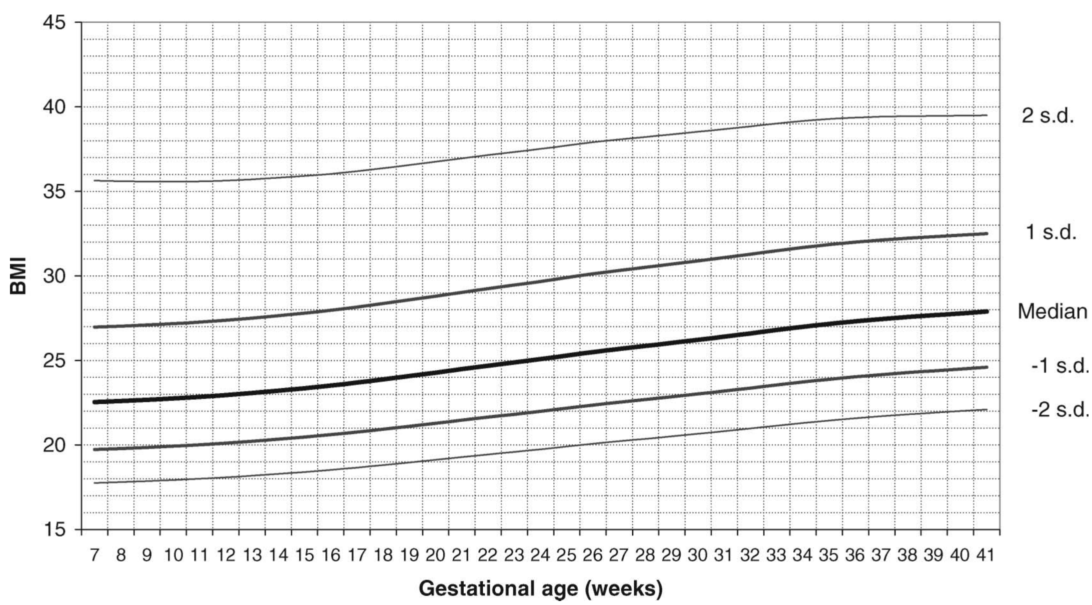

Figure 2. BMI curves by gestational age.

Table IV. LMS parameters and centiles of BMI per gestation week.

Gestation week	BMI							
	L	*M*	*S*	*c* 3	*c* 10	*c* 50	*c* 90	*c* 97
6	−1.96	22.43	0.15	17.75	18.98	22.43	28.80	34.96
7	−1.94	22.45	0.15	17.76	18.99	22.45	28.79	34.86
8	−1.91	22.48	0.15	17.78	19.01	22.48	28.79	34.78
9	−1.89	22.52	0.15	17.81	19.05	22.52	28.82	34.73
10	−1.87	22.59	0.15	17.86	19.10	22.59	28.87	34.71
11	−1.85	22.67	0.15	17.92	19.17	22.67	28.94	34.72
12	−1.83	22.77	0.15	17.99	19.26	22.77	29.05	34.77
13	−1.81	22.90	0.15	18.09	19.36	22.90	29.17	34.85
14	−1.79	23.03	0.15	18.19	19.48	23.03	29.32	34.96
15	−1.77	23.18	0.15	18.30	19.59	23.18	29.47	35.07
16	−1.76	23.33	0.15	18.42	19.72	23.33	29.64	35.21
17	−1.74	23.51	0.15	18.55	19.87	23.51	29.83	35.38
18	−1.72	23.70	0.15	18.70	20.03	23.70	30.04	35.57
19	−1.71	23.89	0.15	18.86	20.20	23.89	30.26	35.77
20	−1.69	24.09	0.15	19.02	20.37	24.09	30.48	35.98
21	−1.68	24.30	0.15	19.18	20.55	24.30	30.72	36.20
22	−1.67	24.51	0.14	19.35	20.73	24.51	30.95	36.42
23	−1.66	24.71	0.14	19.51	20.91	24.71	31.17	36.64
24	−1.65	24.92	0.14	19.68	21.08	24.92	31.40	36.85
25	−1.63	25.13	0.14	19.85	21.27	25.13	31.64	37.07
26	−1.62	25.34	0.14	20.02	21.45	25.34	31.86	37.28
27	−1.61	25.52	0.14	20.17	21.61	25.52	32.05	37.45
28	−1.61	25.69	0.14	20.31	21.76	25.69	32.22	37.60
29	−1.60	25.86	0.14	20.46	21.91	25.86	32.40	37.76
30	−1.59	26.04	0.14	20.61	22.08	26.04	32.59	37.92
31	−1.58	26.23	0.14	20.77	22.25	26.23	32.78	38.10
32	−1.58	26.42	0.14	20.94	22.42	26.42	32.98	38.28
33	−1.57	26.62	0.14	21.11	22.61	26.62	33.19	38.46
34	−1.56	26.81	0.14	21.28	22.78	26.81	33.38	38.62
35	−1.55	26.98	0.14	21.43	22.94	26.98	33.53	38.75
36	−1.54	27.12	0.14	21.56	23.07	27.12	33.66	38.83
37	−1.54	27.25	0.14	21.68	23.20	27.25	33.76	38.88
38	−1.53	27.36	0.14	21.78	23.30	27.36	33.84	38.91
39	−1.52	27.45	0.14	21.87	23.40	27.45	33.90	38.91
40	−1.51	27.54	0.13	21.97	23.49	27.54	33.96	38.92
41	−1.50	27.64	0.13	22.06	23.59	27.64	34.03	38.93
42	−1.49	27.74	0.13	22.16	23.69	27.74	34.09	38.95

Table V. Weight increments by gestational age.

Gestational age (weeks)	Weight increment (kg/month)			
	Women with newborns ≥3000 g		Women with newborns <3000 g	
	Mean	95% CI	Mean	95% CI
<16	1.24	1.12–1.35	0.97	0.79–1.15
16.1–20	2.02	1.91–2.12	1.74	1.55–1.93
20.1–24	2.16*	2.06–2.26	1.74*	1.59–1.89
24.1–28	2.07	1.98–2.17	1.91	1.77–2.05
28.1–32	1.84	1.74–1.94	1.66	1.51–1.80
32.1–36	1.96*	1.86–2.10	1.66*	1.48–1.83
36.1–40	2.26	2.09–2.42	1.94	1.66–2.22

*Significantly different between categories of birth weight.

>+1 SD had a sensitivity of 40.33% and a specificity of 84.92%. These cut-off points had in this case a high specificity and a relatively low sensitivity, with a global efficiency of 77% and 81% for low and high birth weight, respectively.

Discussion

This study presents a new BMI for gestational age chart based on a longitudinal follow-up of a cohort of pregnant women.

Since 1987, there has not been published data related to pregnancy weight gain coming from longitudinal designs in a population of pregnant women from a Latin American country, with the exception of a recent study in Buenos Aires city, which calculated weight gain and BMI on a cohort of 243 pregnant women [18].

This proposal has the relevance of its adequate sample size, its wide country representation, its longitudinal design including the newborn outcomes, and that the parameters of weight gain and

BMI have been generated using the new LMS methodology.

Several limitations must be considered in interpreting the results: a 10.8% of smokers and 7.3% of alcohol consumers have been included in the cohort. Number of cigarettes does not exceed five a day and alcohol intake was less than 20 g of ethanol per day. Although both alcohol intake and smoking during pregnancy are known to have a negative incidence on birth weight, no differences in birth weight (t-test = 0.362, p = 0.717) or in maternal weight gain (t-test = 0.7096, p = 0.478) were found in those women who smoke. There were also no differences in birth weight (t-test = 0.0325, p = 0.974) or in maternal weight gain (t-test = 1.409, p = 0.159) in those who consume small quantities of alcohol.

The proposed charts are based on adult women who had singleton pregnancies, and therefore cannot be applied to adolescent pregnant women or to twin or multiple pregnancies.

Maternal weight gain monitoring is an important component of pre-natal care and in the last years there has been an increased interest worldwide in generating instruments to be used with this purpose [19–22].

Instruments that are currently in use to evaluate maternal weight gain during pregnancy in Latin American countries are the Rosso-Mardones chart [12] and the new proposal from the University of Chile [15]. The Rosso-Mardones chart defines categories of maternal nutritional status based on weight/height expressed either as percentage of standard weight or BMI. Cut-off points were defined considering the risk of adverse neonatal outcomes. This chart has been proposed by the Chilean National Health Service in 1987 and has been used in other countries like Argentina, Brazil, Ecuador, Panama and Uruguay in their pre-natal care programs. Nevertheless, during the last years, there is evidence that the cut-off points proposed by Mardones and Rosso overestimate both under- and over-nutrition. For these reasons, Atalah et al. [15] generated a new proposal to assess the nutritional status of pregnant women. This table was theoretically defined using the criteria of normality and of weight increase from the literature.

The recommendations for weight gain in pregnancy from the IOM [11] are another internationally used instrument to monitor pregnancy weight gain. It has been questioned because expressing weight gain in absolute values does not consider proportionality of weight gain in smaller and taller than average women. Besides, some authors argued that the evidence of benefit from the IOM's recommendations is weak [8,9]. However, recent publications including large samples of pregnant women have demonstrated that gestational weight gain above the IOM's guidelines was associated with multiple adverse neonatal outcomes, whereas gestational weight gain below guidelines was associated with small for gestational aged newborns [23,24].

Nevertheless, in Argentina the IOM's references are applied for nutritional classification of pregnant women in research protocols but are not frequently used during the pre-natal control; the graphical presentation of both Rosso-Mardones and the University of Chile Charts makes them more appropriate for clinical practice.

In a recent study, Ochsenbein-Kölble et al. [25] have generated new reference ranges for weight gain and increase in BMI during pregnancy from a large population of Caucasian, Asian and Black European pregnant women, using also the LMS method. Compared with the Argentine women, Caucasian women in the Ochsenbein-Kölble study were taller (163 cm vs. 159 cm), and presented higher weight gains during the whole pregnancy, mean weight gain observed in their study was 15.5 $\pm$ 5.9 kg compared with 11.9 $\pm$ 4.4 kg in the Argentine women. Median BMI values at Week 10 were 23.5 for the Caucasian European women and 22.6 in the present sample, and at Week 40 they were 28.3 and 27.5, respectively. Their study design was cross-sectional with different sample sizes at particular gestational ages, with subjects concentrated in the first or in the third trimester. In the present study, there are a pretty constant number of measurements of the same subjects along gestation. On the other hand, there could be a selection difference because the subjects in the European study were recruited in a reference University Hospital and the present ones in antenatal clinics; incidence of pathology such as gestational diabetes (12% vs. 1.9%) could be related to this selection bias. Finally, our study attempts to define categories of pregnant women nutritional status related to birth weight.

Compared with the Grandi et al. study [18], our results are quite similar in terms of characteristics of the population and total weight gain. However, the shape of their curve is rather irregular in the higher percentiles, probably because of the small sample size.

The pattern of maternal weight gain observed in this study was similar to other reports, with lower rates of weight gain during the first weeks of gestation, peaking during weeks 20–24 and slowing down at the end of pregnancy [26]. Abrams and Selvin examined the relationship between trimester patterns of maternal weight gain and birth weight, and emphasised that weight gain during the second trimester is particularly related to birth weight. In the present cohort, a significant difference in weight gain patterns at weeks 20–24 is also observed between mothers of neonates with birth weights less than

3000 g and those with $\geq$3000 g. The formers presented an average weight gain of 1.74 kg, compared with 2.16 kg in mothers of infants heavier than 3000 g.

No attempt to extend the follow-up until the post-partum period was made, as it was not considered in the study hypothesis. It would be interesting to evaluate other outcomes besides birth weight, as post-partum weight retention.

Sensitivity of maternal BMI for the prediction of low birth weight was relatively low in this study. That is not surprising because it has to be considered that maternal weight gain is not the main determinant of birth weight, and that other proposed instruments (like Rosso-Mardones or Atalah Charts) also have low sensitivity.

The BMI curves proposed in the present study need further validation to find out the best cut-off points for risk prediction in populations with different prevalences of low or high birth weights. Nevertheless, this instrument represents an improvement for pre-natal care monitoring at the national level.

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References

1. Ramakrishnan U. Nutrition and low birth weight: From research to practice. Am J Clin Nutr 2004;79:17–21.
2. Hulsey TC, Neal D, Bondo SC, Hulsey T, Newman R. Maternal pre-pregnant body mass index and weight gain related to low birth weight in South Carolina. South Med J 2005;98:411–415.
3. Sekiya N, Anai T, Matsubara M, Miyazaki F. Maternal weight gain rate in the second trimester are associated with birth weight and length of gestation. Gynecol Obstet Invest 2007;63:45–48.
4. Nahar S, Mascie-Taylor CG, Begum HA. Maternal anthropometry as a predictor of birth weight. Public Health Nutr 2007;10:965–970.
5. Kabiru W, Raynor BD. Obstetric outcomes associated with increase in BMI category during pregnancy. Am J Obstet Gynecol 2004;191:928–932.
6. Stotland NE, Hopkins LM, Caughey AB. Gestational weight gain, macrosomia, and risk of cesarean birth in nondiabetic nulliparas. Obstet Gynecol 2004;104:671–677.
7. Nohr EA, Bech BH, Davies MJ, Frydenberg M, Henriksen TB, Olsen J. Prepregnancy obesity and fetal death: A study within the Danish national birth cohort. Obstet Gynecol 2005;106:250–259.
8. Abrams B, Altman S, Pickett KE. Pregnancy weight gain: Still controversial. Am J Clin Nutr 2000;71:1233S–1241S.
9. Johnson JW, Yancey MK. A critique of the new recommendations for weight gain in pregnancy. Am J Obstet Gynecol 1995;174:254–258.
10. ACOG. Nutrition during pregnancy. Technical Bulletin Number 179. Int J Gynaecol Obstet 1993;43:67–74.
11. Subcommitee on Nutritional Status and Weight Gain During Pregnancy. Food and Nutrition Board, National Academy of Sciences. Washington DC: National Academy Press; 1990.
12. Mardones F, Rosso P. Design of a weight gain chart for pregnant women. Rev Méd Chile 1997;125:1437–1448.
13. Mardones F, Rosso P. A weight gain chart for pregnant women designed in Chile. Matern Child Nutr 2004;1: 77–90.
14. Atalah E, Castillo C, Gomez C, Mateluna A, Urteaga C, Castro R, Raimallo R. Undernutrition in pregnant women: An overestimated problem? Rev Méd Chil 1995;123:1531–1538.
15. Atalah E, Castillo C, Castro R, Aldea A. Proposal for a new standard for the nutritional assessment of pregnant women. Rev Méd Chil 1997;125:1429–1436.
16. World Health Organization. Physical status: the use and interpretation of anthropometry. Report of a WHO Expert Committee. Technical Report Series No. 854. Geneva, 1995.
17. Cole TJ, Green PJ. Smoothing reference centile curves: The LMS method and penalized likelihood. Stat Med 1992; 11:1305–1319.
18. Grandi C, Luchtenberg G, Sola H. Evaluación nutricional durante el embarazo. Nuevo estándar. Medicina (B Aires) 2007;67:677–684.
19. Voigt M, Straube S, Schmidt P, von Steinburg PS, Schneider KT. Standard values for the weight gain in pregnancy according to maternal height and weight. Z Geburtshilfe Neonatol 2007;211:191–203.
20. Mahaba HM, Ismail NA, El Teheiwy MM, El-Goewily MM, Ramadan MS. Development of weight gain charts for healthy Egyptian pregnant women. J Egypt Public Health Assoc 2001;76:369–391.
21. Wong W, Tang NL, Lau TK, Wong TW. A new recommendation for maternal weight gain in Chinese women. J Am Diet Assoc 2000;100:791–796.
22. Takimoto H, Sugiyama T, Fukuoka H, Kato N, Yoshiike N. Maternal weight gain ranges for optimal fetal growth in Japanese women. Int J Gynaecol Obstet 2006;92:272–278.
23. Devader SR, Neeley HL, Myles TD, Leet TL. Evaluation of gestational weight gain guidelines for women with normal prepregnancy body mass index. Obstet Gynecol 2007;110:745–751.
24. Brawarsky P, Stotland NE, Jackson RA, Fuentes-Afflick E, Escobar GJ, Rubashkin N, Haas JS. Pre-pregnancy and pregnancy-related factors and the risk of excessive or inadequate gestational weight gain. Int J Gynaecol Obstet 2005;91:125–131.
25. Ochsenbein-Kölble N, Roos M, Gasser T, Zimmermann R. Cross-sectional study of weight gain and increase in BMI throughout pregnancy. Eur J Obstet Gynecol Reprod Biol 2007;130:180–186.
26. Abrams B, Selvin S. Maternal weight gain pattern and birth weight. Obstet Gynecol 1995;86:163–169.